Agenda

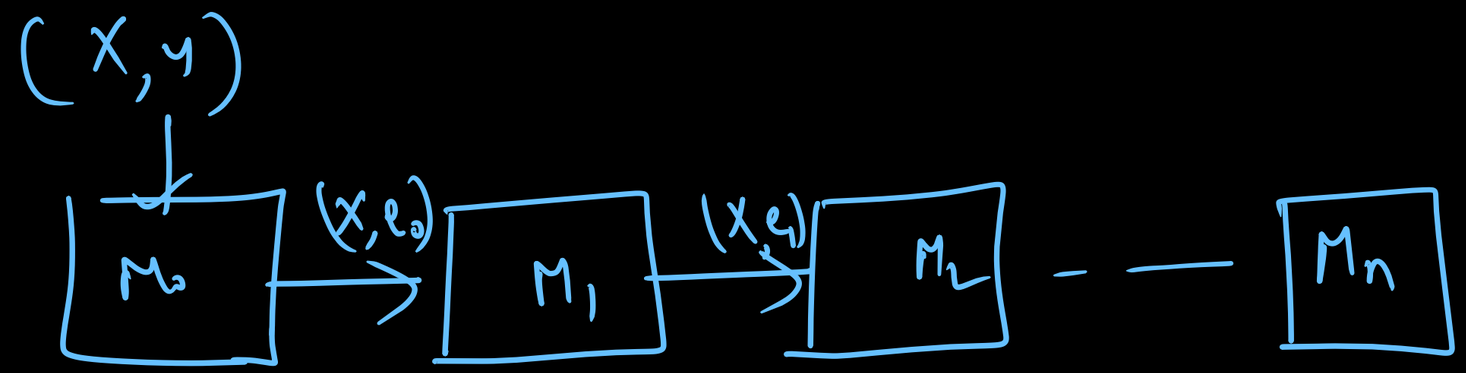
- → XGBoost
- → LightGBM
- → Stacking
- → Cascading

XG Boost

↳ Extreme Gradient Boosting

↳ Faster for training

scikit GBM	XG Boost
→ Pure Python	→ C++ backend ↓ Fast
→ Parallelism is limited	→ More Parallelism
→ No GPU support	→ GPU support



1. Parallelization of feature Selection

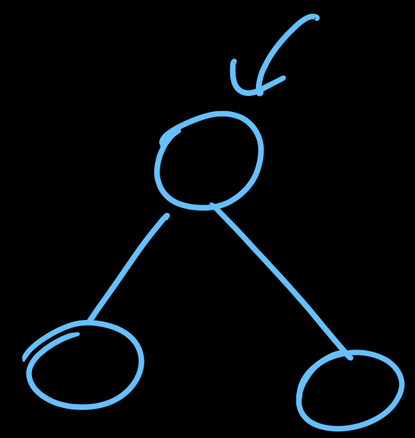
$f_1 \rightarrow IG_1$

$f_2 \rightarrow IG_2$

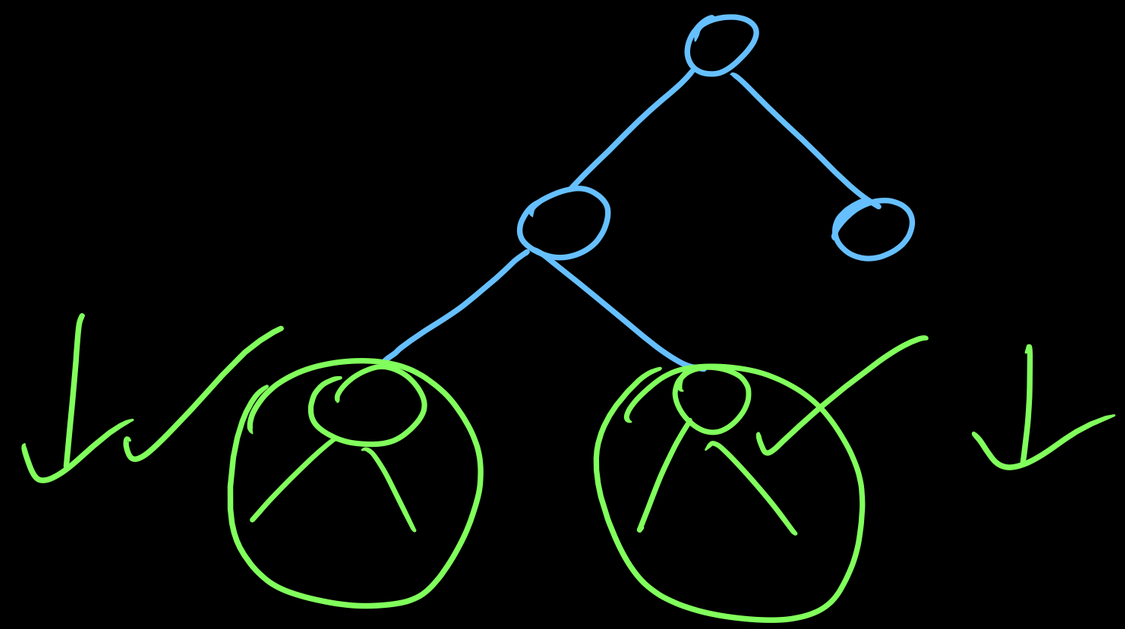
$\vdots$

$f_m \rightarrow IG_m$

→ Compute in parallel



② Parallelization in building DT



③ Optimizing threshold in numerical feature

1. Sort in ascending
2. Take each unique value as threshold
3. Calculate IG on each threshold

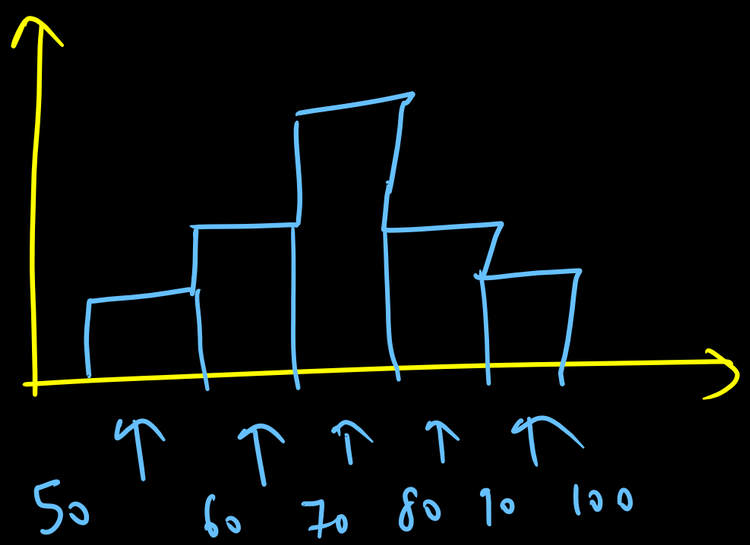
100 bins

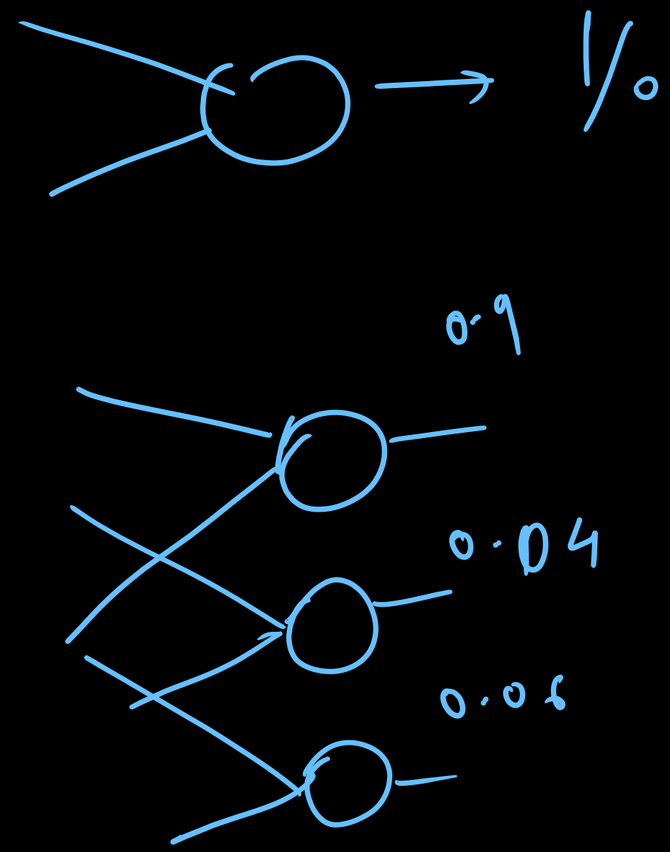
2	10
3	10
11	20
20	20
25	30
29	30

$<= 10$

$11 < \quad < 20$

$20 < \quad \leq 30$




1/0
0.9
0.04
0.06

Light GBM

↳ lite GBM

1. Gradient Based One Sided Sampling
2. Exclusive Feature Bundling.

$M_0 \rightarrow M_1 \longrightarrow \underline{\underline{M_2}} \longrightarrow M_3 \longrightarrow M_4 \longrightarrow M_5$

20M

$M_3 \rightarrow err \sim 0$

16M

15M

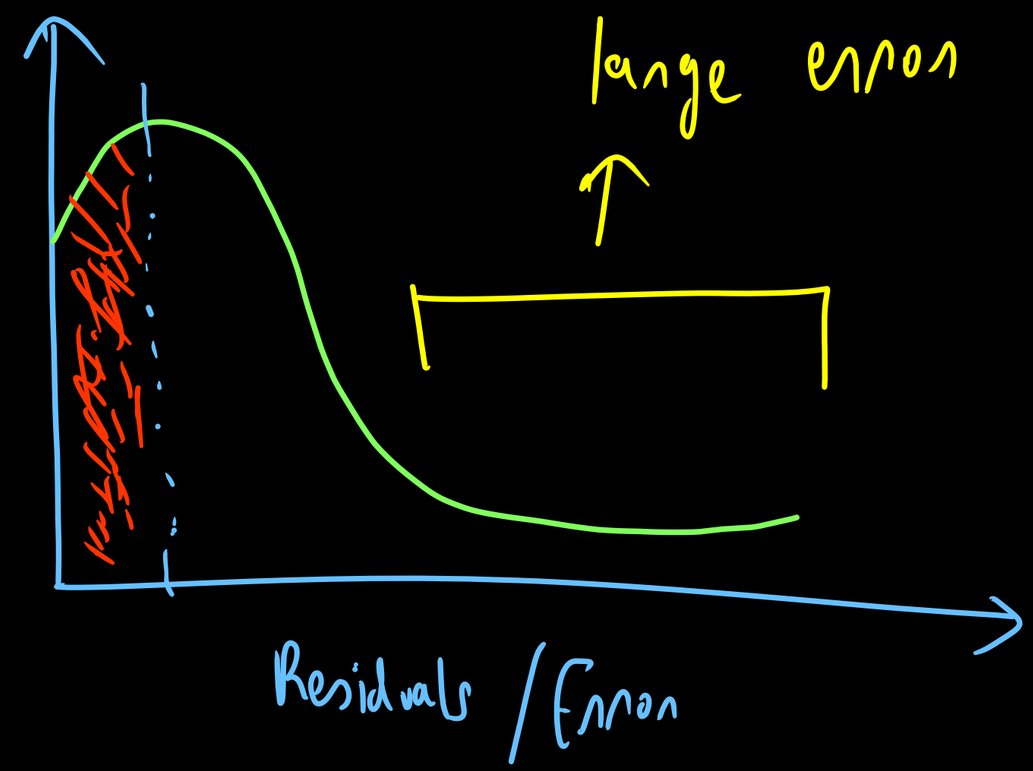
14M

0.0

5×10^{-3}

2×10^{-4}

0.0056



Exclusive Feature Bundling (EFB)

- Reduces the no of features
- Applied to any sparse features which are mutually exclusive.

F1	F2	F3
1	0	0
0	1	0
0	0	1
1	0	0
0	1	0

→

F_bundle
1
2
3
1
2

→ Memory Saving

→ Faster training due to less feature

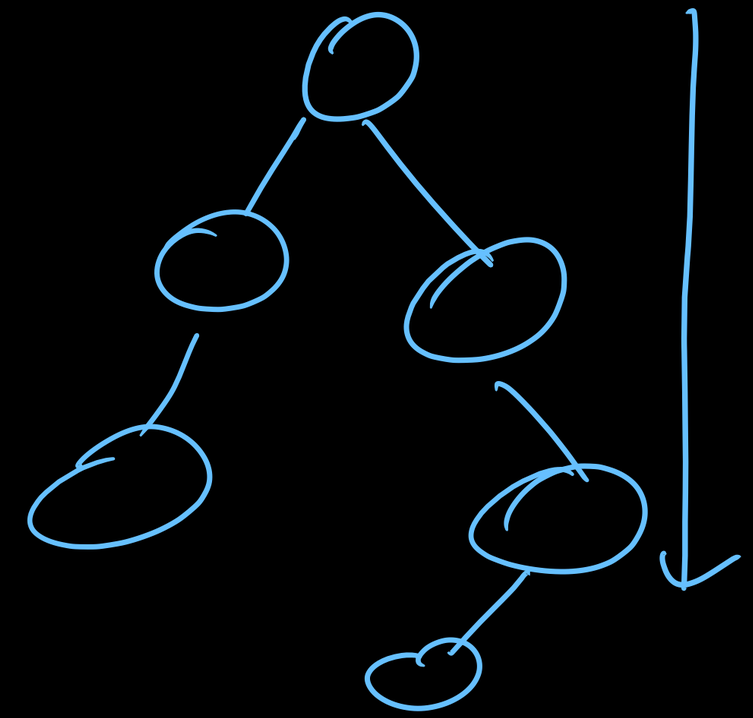
F1	F2	F3	F4	F_bundle
0	5	0	2	5
0	0	0	0	0
3	0	0	3	3
0	0	2	2	2
0	0	2		2
		0		0
		0		0
		2		2

1 2 3

{ 2 : f3
5 : f2
3 : f1 }

→ Graph based

DT1 DT2



→ Break until 22:28 PM

→ One hot encoding

- ✓ → XG Boost
- ✓ → Light GBM
- ✓ → Cat Boost
- → RF

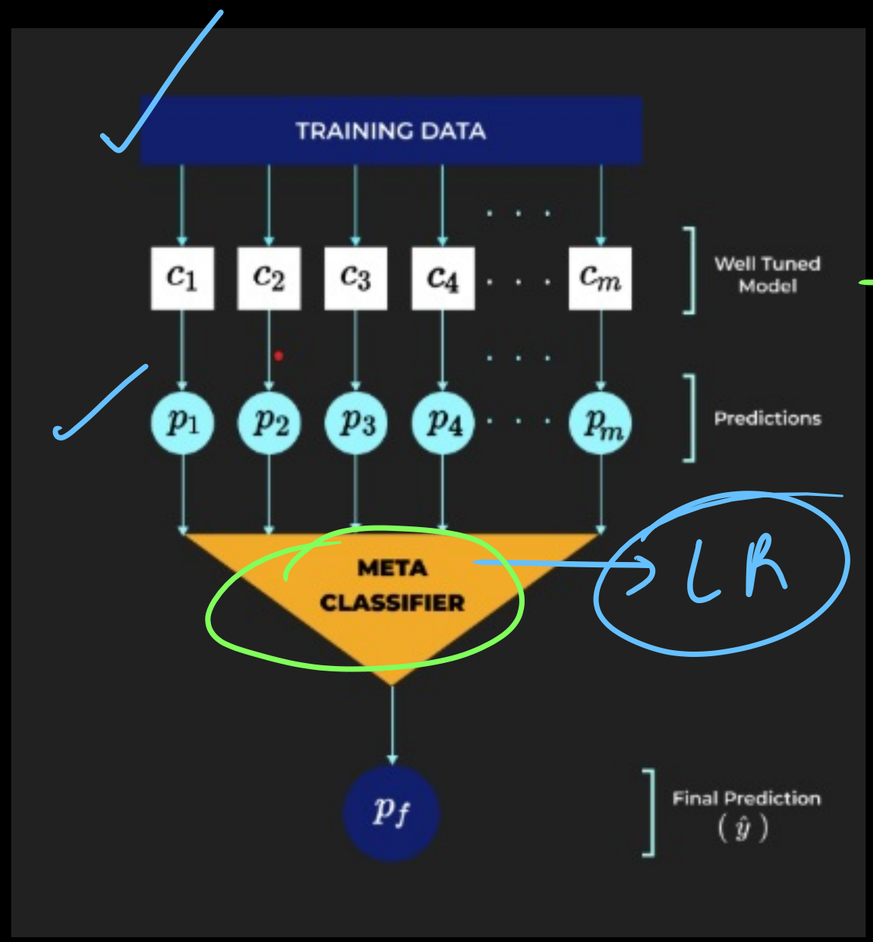
→ Can handle categorical data

Stacking

→ Kaggle

0.8432 → 1st rank

0.840 → 5th rank



→ KNN, LR, DT

LR

Bagging

$\hat{y}_1$	$\hat{y}_2$	$\hat{y}_3$	$\hat{y}$
1	0	1	1
0	1	0	0
1	0	1	1

$P(y=1/x) > 0.5 \rightarrow 1$

0

$\hat{y}_1 \quad \hat{y}_2 \quad \hat{y}_3 \rightarrow \boxed{LR} \rightarrow$

			$\hat{y}$
0.9	0.05	0.85	1
0.1	0.95	0.01	1
0.85	0.05	0.90	1

1

0

LR

Cascading

Deeper model (NN)

$D \rightarrow \boxed{M_1} \rightarrow \boxed{M_2} \rightarrow \boxed{M_3} - - \boxed{M_n}$

$D - D_1$

$D - D_1 - D_2$

$P(y=1/x)$ $< 0.1 \; (D_1)$ → Not Fraud

$P(y=1/x)$ $< 0.1 \; (D_2)$ → Not Fraud

1 → Fraud

0 → Not Fraud

0.95

0.2 — 0.8

0.1

0.11

0.15

0.2

Bagging

→ Aggregation

→ R.S + C.S

→ Homogenous (DT)

Stacking

→ Classifier/Regressor

→ Entire Training data

→ Hetenogenous (LR, DT, KNN)